ARL DSRC TASK ORDER FOR HITS-U-III

GENERAL DESCRIPTION:

ARL administers and operates one of the five DoD HPC DSRCs established under the auspices of the DoD HPCMP. Located at Aberdeen Proving Ground in Maryland, ARL DSRC utilizes world-class HPCs, cutting-edge applications, and expert staff scientists to help the United States maintain its technological and military supremacy. The ARL DSRC offers a full spectrum of computational capabilities for the Department of Defense (DoD) Science and Technology and Test and Evaluation communities, including powerful parallel processors, high-speed networks, applications software, and comprehensive storage solutions. The ARL DSRC also provides operational and administrative support for select HPC class machines for other customers.

The ARL DSRC also operates a portion of the Data Analysis and Visualization Center (DAV-C) in coordination with the ERDC DSRC, providing scientific visualization, conceptual animation and video production support to DoD HPCMP customers.

PWS 3.0 Requirements:

3.1 ARL DSRC OPERATIONS SUPPORT

This PWS element includes all minimal support required for basic operations of the ARL DSRC to be ready to deliver HPC capabilities and assist users. The requirements are somewhat independent of the number of HPC systems operating and HPC users being supported. The requirements are inclusive for a DSRC facility with appropriate operational elements of total electrical power and mechanical cooling capacity (not tied specifically to any HPC systems but representative of overall total capacity), basic DREN and SDREN networking enclaves with various operational workstations and servers, safety and security programs, and business processes such as procurement, asset/inventory control, change management, and configuration management.

Baseline ARL DSRC operations support will ensure the following:

- Operation and delivery of HPC capabilities 24 hours/day, 7 days/week, 365 days/year (includes government holidays, although reduced staff expected during government holidays)
- Facilities, support infrastructure, and non-HPC systems remain operational and reliable
- Cybersecurity support is continuous and consistent with effective risk assessment and management so that network enclaves maintain appropriate authorities to operate
- Hardware and software maintenance agreements are up to date

DRAFT

- Configurations are managed effectively
- User engagement activities including allocations, local marketing & local A/V support.

3.1.1 Operations Monitoring

The Contractor shall provide support to monitor the ARL DSRC facilities, infrastructure elements, and HPC systems for 24x7 operations. This includes all unclassified and classified areas in multiple buildings.

3.1.1.1 Perform physical security functions in accordance with secure facilities operating policies and procedures; to include open/close duties for the rooms, alarm/disarm Intrusion Detection Systems (IDS), and physical checks of safes in classified areas. The contractor will not be responsible for the recall roster.

3.1.1.2 Operations Monitoring is required at 6791 Aberdeen Blvd., Building 120, Aberdeen Proving Ground, Maryland 21005 and 6391 Rodman Road, Building 3511, Aberdeen Proving Ground, Maryland 21005.

3.1.2 Facilities Engineering Support

The Contractor shall provide support to ARL DSRC efforts related but not limited to minor building alterations, environmental monitoring, electrical power and backup subsystems, and mechanical support systems. Services include planning through engineering designs and assessments, as well as other functions needed to execute facility projects.

3.1.2.1 Facilities Engineering Support is required at 6791 Aberdeen Blvd., Building 120, Aberdeen Proving Ground, Maryland 21005 and 6391 Rodman Road, Building 3511, Aberdeen Proving Ground, Maryland 21005.

3.1.2.2 Operate sit down forklift trucks to support infrastructure upgrades; forklift operators will need to have a certificate on file with COTR approval and will be required to have and properly use personal protective equipment (e.g., hard hat, steel toed shoes, and eye protection). Contractors found to be operating unsafely will have privileges revoked until completing a new certification.

3.1.2.3 Provide dedicated support for Building 120 and Building 3511 for operational computer rooms, continued build out of computer rooms in Module 1 and work with local base Civil Engineering, Army Corp of Engineers, vendors, and/or other government stakeholders on Module 2 buildout phases.

DRAFT

3.1.3 Cybersecurity Support for DSRC Operations

The baseline operational DREN, SDREN and/or Above-Secret network enclaves in the ARL DSRC require authorization to operate and connect to the wide-area networks. The Contractor shall assist the government cybersecurity staff in attaining and sustaining the appropriate authorizations to operate. Cybersecurity requirements related to DSRC operations are identified in HITS U III PWS General Scope of Requirements sections 3.7 and 16.8. Cybersecurity support in classified enclaves other than SDREN will be addressed in individual DSRC tasks.

3.1.4 System Administration

For baseline ARL DSRC operations, there are a number of non-HPC devices such as workstations, servers, repositories, and virtual communication equipment. The Contractor shall provide system administration support in keeping these systems operating reliably and securely.

3.1.4.1 Provide cybersecurity RMF support and compliance on a continuous, risk-management based approach and during inspections to achieve and maintain Assess and Authorization or Assess-Only Authority to Operate (ATO) status.

3.1.4.2 Create and maintain the documentation of all systems, configurations, concept of operations, and details on the authentication to systems which are required to support RMF effort.

3.1.4.3 Maintain installed software and software licenses, except for those directly supported by the OEMs.

3.1.4.4 Maintain classified and unclassified data storage systems.

3.1.4.5 Provide Tier II support to users for all HPC systems.

3.1.4.6 Administer, manage and optimize Linux systems for use within the ARL DSRC at unclassified and classified levels.

3.1.5 Network Engineering and Administration

For baseline DSRC operations, the DREN, SDREN and/or Above-Secret enclaves require engineering and system administration support for the various equipment items such as routers and switches. The Contractor shall provide network engineering and system administration for the network devices. In addition, in accordance with DoD HPCMP direction and JFHQ-DODIN Comply-to-Connect (C2C) Concept of Operations instructions, the Contractor shall administer/maintain the necessary hardware and

DRAFT

software (currently Forescout) at the ARL DSRC to continue enforcing C2C compliance on the DREN and SDREN.

3.1.5.1 The Contractor shall provide complete cybersecurity support for network devices ensuring they remain in compliance at all times coordinating (with all monitoring activities).

3.1.5.2 Maintain the network topology, intersystem connection, and data flow diagrams which support RMF activities and update at least semi-annually, or within 10 days of government direction.

3.1.5.3 Coordinate with the ARL Network Team for items listed but not limited to: DREN/SDREN bandwidth and connections to DSRC network devices, WiFi and hardwire network connections within Building 120 & Building 3511 as applicable.

3.1.6 Database Administration

The ARL DSRC maintains databases for baseline operations support. This includes but is not limited to environmental monitoring data, service tickets, maintenance logs, etc. The Contractor shall provide database administration support for these systems.

3.1.6.1 The contractor shall perform database administration and maintain security posture on databases local to the ARL DSRC that require skills beyond those required for System Administration.

3.1.6.2 The Contractor shall develop and sustain local database applications and tools utilized by the AFRL DSRC.

3.1.6.3 The Contractor shall develop and support database solutions that interface with multiple applications in support of authentication and authorization services.

3.1.7 Software Development

The Contractor shall develop and support execution of specialized software used in ARL DSRC operations. This includes but is not limited to applications to monitor facility environmental controls and automate operational processes.

3.1.7.1 Software in this context includes, but is not limited to, job scheduling/workload management, container orchestration, and virtualization software which may or may not be provided on HPC systems by the vendor or original equipment manufacturer (OEM), as well as user authentication software.

3.1.7.2 Integrate, configure, and maintain the Workload Characterization (WC) software, utilities/tools utilized to gather metrics, reporting tools such as

DRAFT

dashboards and other data products, stored data, and user profiles/roles on the WC system.

3.1.7.3 Develop, maintain and support software in accordance with industry best processes and practices known to reduce cost, schedule and performance risk, and in accordance with cybersecurity software development policy.

3.1.8 Maintenance Support

The Contractor shall provide support for ARL DSRC hardware and software license and maintenance agreements related to facility equipment,supporting infrastructure (networks, non-HPC servers, etc.), and HPCs (allocated and unallocated)used in routine operations. Note: Portions of the hardware and software maintenance are currently performed via OEM or third-party maintenance personnel under contracts acquired directly by the Government.

3.1.8.1 Maintenance Support as outlined above is inclusive of both Building 120 & Building 3511 as applicable.

3.1.9 Inventory Management Support

The Contractor shall assist the government in identifying and documenting equipment inventory for asset management. This contract does not handle Government Furnished Property (GFP); however, the Contractor shall provide support to the ARL DSRC in receiving, identifying, barcoding and documenting equipment inventory for asset management.

3.1.9.1 Support the Government by maintaining an inventory of all HPCMP and DSRC equipment in government-required formats/systems and provide inventory reports.

3.1.9.2 Support the Government in decommissioning and disposing of communications-computer systems no longer needed or serviceable in accordance with Government requirements. Update hardware and software inventory, as well as applicable infrastructure diagrams and system security plans.

3.1.10 Configuration and Change Management

Various system and subsystem configurations often change within the ARL DSRC and effective management processes are executed. The Contractor shall support the configuration and change management processes by assisting in identifying and documenting configurations and associated changes.

DRAFT

3.1.10.1 The Contractor shall develop, sustain and use an ARL DSRC internal change management system backend database and front-end interface following a software development lifecycle and in accordance with software development security policy.

3.1.11 Procurement Support

The ARL DSRC has the ability to purchase various items needed to support routine operations. This includes, but is not limited to, office supplies and equipment, facility infrastructure items, and HPC infrastructure equipment, as well as hardware/software maintenance. The Contractor shall provide procurement/purchasing support for any ARL DSRC requirements.

3.1.11.1 The contractor shall develop, sustain and use an ARL DSRC internal procurement management system backend database and front-end interface, following a software development lifecycle and in accordance with software development security policy.

3.1.12 Continuity of Operations

Technical support is required to reconstitute critical operations, defined by the government, at alternative location(s) in the event of a disaster that causes the shutdown of ARL DSRC operations at the primary site. The Contractor shall assist in establishing continuity of operations and disaster recovery for such events and shall support any exercises/simulations.

3.1.13 Technical Project Support

The ARL DSRC may execute studies and projects related (but not limited) to facility upgrades, infrastructure changes, HPC capacity increases, and HPC system installation and integration. The Contractor shall provide technical support to ARL DSRC studies and projects, whether at the local or enterprise level. This support shall include technology planning and assessments and engineering support during project execution.

3.1.14 User Engagement

The ARL DSRC provides support to the Army Allocation Officer to allocate Army HPC hours and maintain accurate Army usage reports. In addition, the contractor shall provide support in generating materials showcasing the ARL DSRC to perspective users, current users and DoD Leadership. Additionally, the contractor shall provide Audio/Video support to the ARL DSRC in support of daily operations.

3.2 DELIVER UNCLASSIFIED HPC CAPABILITIES AT THE ARL DSRC

DRAFT

Delivering HPC capabilities includes requirements in direct support of installing, integrating, and operating HPC systems for the HPCMP. The baseline scope in this PWS section includes allocated, shared unclassified HPC systems as well as shared special-purpose unclassified systems. This PWS element also includes the HPC-related infrastructure that ties directly to delivering unclassified HPC capabilities, such as data movement, data storage (mid-tier and archive), and user interfaces/portals to unclassified HPC systems. The ARL DSRC currently hosts two (2) allocated, shared unclassified HPC systems.

3.2.1 Infrastructure Support for Unclassified HPC Systems

The Contractor shall provide technical support to include, but not be limited to, system engineering, system integration, system administration, and database administration for HPC-related infrastructure directly tied to delivering unclassified HPC capabilities. This includes (but is not limited to) HPC data movement and storage systems, portals, servers, routers, and switches that have direct relationships/connections to HPC systems. This does not include those systems or devices that are part of the ARL DSRC baseline operations identified in paragraph 3.1 and associated subparagraphs.

3.2.2 Unclassified HPC System Installation and Integration

This requirement includes preparation of the ARL DSRC facilities for installing, integrating, accepting, and operating unclassified HPC systems. The Contractor shall provide engineering support for connecting the unclassified HPC systems to the ARL DSRC infrastructure, both facilities-related and HPC-delivery-related.

3.2.3 Application Management Support for Unclassified HPC

The HPC original equipment manufacturers (OEMs) typically provide levels of support, maintenance, and warranties for hardware and software delivered with an operational HPC system. For those software components that are to be installed and/or modified to support HPCMP workloads/use cases, the Contractor shall install, modify, and manage those applications for unclassified HPC system utility. This includes (but is not limited to) job schedulers, reservation/dedicated partition services, data management/movement services, data visualizers, HPC applications, and user interfaces/portals.

3.2.4 Cybersecurity Support for Unclassified HPC Systems

This requirement is related to cybersecurity support for unclassified HPC needed in addition to the ARL DSRC operations cybersecurity support identified in paragraph 3.1.3. The Contractor shall provide this additional cybersecurity support to include (but not be limited to) HPC infrastructure and HPC systems. Cybersecurity requirements related to unclassified HPC systems are identified in HITS U III PWS General Scope of Requirements section 3.7.

DRAFT

3.2.5 Unclassified HPC System Support

Unclassified HPC systems may be managed and administered by OEMs, third-party entities, government personnel, or the Contractor. The Contractor shall provide additional unclassified HPC system administration support including (but not limited to) job scheduler configuration, regular backups of critical data, assistance with troubleshooting issues. Additionally, the contractor will be responsible for the planning, implementation, maintenance and sustainment of the Persistent Services Framework (PSF), a continually running application that provides extended functionality in support of HPC user workloads, on all allocated unclassified HPC systems at the ARL DSRC.

3.2.6 Additional unclassified HPC requirements

To be determined.

3.3 DELIVER CLASSIFIED HPC CAPABILITIES AT THE ARL DSRC

Delivering HPC capabilities includes requirements in direct support of installing, integrating, and operating HPC systems for the HPCMP. The baseline scope in this PWS section includes allocated, shared classified HPC systems as well as shared special-purpose classified HPC systems. This PWS element also includes the HPC-related infrastructure that ties directly to delivering classified HPC capabilities, such as data movement, data storage (mid-tier and archive), and user interfaces/portals to classified HPC systems. The ARL DSRC currently hosts four (4) allocated, shared classified HPC systems.

3.3.1 Infrastructure Support for Classified HPC

The Contractor shall provide technical support to include, but not be limited to, system engineering, system integration, system administration, and database administration for HPC-related infrastructure directly tied to delivering classified HPC capabilities. This includes (but is not limited to) HPC data movement and storage systems, portals, servers, routers, and switches that have direct relationships/connections to HPC systems. This does not include those systems or devices that are part of the ARL DSRC baseline operations identified in paragraph 3.1 and associated subparagraphs.

3.3.2 Classified HPC System Installation and Integration

This requirement includes preparation of the ARL DSRC facilities for installing, integrating, accepting, and operating classified HPC systems. The Contractor shall provide engineering support for connecting the classified HPC systems to the DSRC infrastructure, both facilities-related and HPC-delivery-related.

3.3.3 Application Management Support for Classified HPC

DRAFT

The HPC original equipment manufacturers (OEMs) typically provide levels of support, maintenance, and warranties for hardware and software delivered with an operational HPC system. For those software components that are to be installed and/or modified to support HPCMP workloads/use cases, the Contractor shall install, modify, and manage those applications for classified HPC system utility. This includes (but is not limited to) job schedulers, reservation/dedicated partition services, data management/movement services, data visualizers, HPC applications, and user interfaces/portals.

3.3.4 Cybersecurity Support for Classified HPC Systems

This requirement is related to cybersecurity support for classified HPC needed in addition to the ARL DSRC operations cybersecurity support identified in paragraph 3.1.3. The Contractor shall provide this additional cybersecurity support to include (but not be limited to) HPC infrastructure and HPC systems. Cybersecurity requirements related to classified HPC systems are identified in HITS U III PWS General Scope of Requirements section 3.7.

3.3.5 Classified HPC System Support

Classified HPC systems may be managed and administered by OEMs, third-party entities, government personnel, or the Contractor. The Contractor shall provide additional classified HPC system administration support including (but not limited to) job scheduler configuration, regular backups of critical data, assistance with troubleshooting issues. Additionally, the contractor will be responsible for the planning, implementation, maintenance and sustainment of the Persistent Services Framework (PSF), a continually running application that provides extended functionality in support of HPC user workloads, on all allocated classified HPC systems at the ARL DSRC.

3.3.6 Additional classified HPC requirements

To be determined.

DRAFT

3.4 DELIVER SPECIALIZED HPC CAPABILITIES AT THE ARL DSRC

Delivering HPC capabilities includes requirements in direct support of installing, integrating, and operating HPC systems for the HPCMP. The baseline scope in this PWS section includes specialized HPC systems used to explore new technologies, alternative service delivery models, or emerging non-traditional workloads/use cases. These systems are typically not shared or allocated the same as the systems in PWS paragraphs 3.2 and 3.3, and can be hosted in various environments (unclassified, classified, isolated, standalone). This PWS element also includes the HPC-related infrastructure that ties directly to delivering specialized HPC capabilities, such as data movement, data storage (mid-tier and archive), and user interfaces/portals to specialized HPC systems.

3.4.1 Infrastructure Support for Specialized HPC

The Contractor shall provide technical support to include, but not be limited to, system engineering, system integration, system administration, and database administration for HPC-related infrastructure directly tied to delivering specialized HPC capabilities. This includes (but is not limited to) HPC data movement and storage systems, portals, servers, routers, and switches that have direct relationships/connections to HPC systems. This does not include those systems or devices that are part of the ARL DSRC baseline operations identified in paragraph 3.1 and associated subparagraphs.

3.4.2 Specialized HPC System Installation and Integration

This requirement includes preparation of the ARL DSRC facilities for installing, integrating, accepting, and operating specialized HPC systems. The Contractor shall provide engineering support for connecting the specialized HPC systems to the ARL DSRC infrastructure, both facilities-related and HPC-delivery-related.

3.4.3 Application Management Support for Specialized HPC

The HPC original equipment manufacturers (OEMs) typically provide levels of support, maintenance, and warranties for hardware and software delivered with an operational HPC system. For those software components that are to be installed and/or modified to support HPCMP workloads/use cases, the Contractor shall install, modify, and manage those applications for specialized HPC system utility. This includes (but is not limited to) job schedulers, reservation/dedicated partition services, data management/movement services, data visualizers, HPC applications, and user interfaces/portals.

3.4.4 Cybersecurity Support for Specialized HPC Systems

This requirement is related to cybersecurity support for specialized HPC needed in addition to the ARL DSRC operations cybersecurity support identified in paragraph 3.1.3. The Contractor shall provide this additional cybersecurity support to include (but not be limited to) HPC infrastructure and HPC systems. Cybersecurity requirements related to Specialized HPC systems are identified in HITS U III PWS General Scope of Requirements section 3.7.

3.4.5 Specialized HPC System Support

Specialized HPC systems may be managed and administered by OEMs, third-party entities, government personnel, or the Contractor. The Contractor shall provide additional specialized HPC system administration support including (but not limited to) job scheduler configuration, regular backups of critical data, assistance with troubleshooting issues. Additionally, the contractor will be responsible for the planning, implementation, maintenance and sustainment of the Persistent Services Framework

DRAFT

(PSF), a continually running application that provides extended functionality in support of HPC user workloads, on specialized HPC systems at the ARL DSRC.

3.4.6 Additional specialized HPC requirements

To be determined.

3.5 ASSIST HPC USERS AT THE ARL DSRC

This PWS element includes all baseline support required for assisting HPC users. The requirements relate to an enterprise HPC Help Desk basic function hosted by a single DSRC and higher levels of more involved support from experts at the ARL DSRC. Also included is an enterprise data analysis and visualization requirement to assist HPC users and HPCMP staff in generating graphical representations of HPC results. Another requirement is included to support the development and maintenance of user account authentication methods for secure access to HPC systems. As most user assistance requirements are executed with a Centers enterprise perspective, they may be met by contractor staff located at the ARL DSRC, as well as remotely.

3.5.1 Basic HPC Help Desk

The Centers component requires an enterprise-level basic Help Desk function to support all HPC users. ARL DSRC help desk requirements include, but are not limited to, user account setup support and troubleshooting, basic HPC system information, and limited HPC application support. The contractor shall provide HPC customer support services (Monday-Friday, 8:00 am to 4:30 pm EST, 8.5 hours each day) to local and remote users of ARL DSRC unclassified and classified resources.

3.5.2 ARL DSRC Help Desk Liaisons

The Contractor shall provide support staff at the ARL DSRC to follow up on service requests issued by the Basic HPC Help Desk in PWS paragraph 3.5.1. These local ARL DSRC support staff are required to be familiar with the ARL DSRC's HPC systems, infrastructure, and other support elements in order to address each Help Desk service request sent for follow-up. Support is required for both unclassified and classified HPC capabilities at the ARL DSRC.

> 3.5.2.1 ARL DSRC support staff will also be available to answer local ARL DSRC Help Desk phone calls, monitor and respond to ARL DSRC Help Desk unclassified and classified emails during the service times outlined in the HITS U III PWS General Scope of Requirements section 1.5.13.
>
> 3.5.2.2 ARL DSRC support staff will provide support for local classified HPC systems via a local ticketing system (outside of the enterprise HPC Help Desk)

DRAFT

with the requirement to create, monitor and follow up on tickets timely as identified in HITS U III PWS General Scope of Requirements section 4.0.

3.5.3 Higher Level User Support

The Contractor shall provide additional ARL DSRC user support (above the basic level) to address areas such as, but not limited to, HPC application software installation and software container development and support. This level of support typically requires up to thirty (30) days for complex issues. Other requests for support that involve other components within the HPCMP or third parties such as HPC vendors shall be transferred appropriately.

3.5.4 HPC Application Management and Support

The Contractor shall manage and support all HPC enterprise software applications that are accessible to ARL DSRC users for their workflows/use cases. This includes government-owned, commercial, and open-source software installed (by the Centers component) and operating on ARL DSRC HPC systems (both unclassified and classified), as well as license servers and managers.

DRAFT

DIRECTIVES:

The following directives are mandatory and shall be complied with:

ARL-MEMO-25-72	Use of Portable Electronic Devices within ARL, 30 January 2006.
ARL-MEMO-25-78	ARL Personally Identifiable Information Breach Program, 18 November 2008.
ARL-MEMO-380-7	Operational Security (OSPEC) Policy, 21 December 2023.
ARL-R-385-10	ARL Safety Program, 23 June 2023.
ARL-MEMO-380-4	Emergency Plan for Classified Information, 27 May 2007.

The following directives are provided as guidance:

AR 25-1	Department of the Army Information Technology, 15 July 2019.
AR 25-2	Department of the Army Cybersecurity, 04 April 2019.
AR 190-13	Department of the Army Physical Security Program, 27 June 2019.
AR 380-5	Department of the Army Information Security Program, 25 March 2022.
AR 380-67	Department of the Army Personnel Security Program, 24 January 2014.
AR 735-5	Department of the Army Policies and Procedures for Property Accountability, 10 April 2024.
AR 380-10	Department of the Army Foreign Disclosure and Contacts with Foreign Representatives, 13 July 2015.
AR 380-381	Department of the Army Special Access Programs (SAPs) and Sensitive Activities, 21 April 2004.
AR 380-53	Department of the Army Communications and Security Monitoring, 21 April 2004.
AR 380-49	Department of the Army Industrial Security Program, 20 March 2013.
PAM 25-2-1	Department of the Army Cross Domain Solution and Data Transfer Management, 12 April 2019.
PAM 25-2-2	Department of the Army Cybersecurity Tools Unified Capabilities Approved Products List Process, 8 April 2019.
PAM 25-2-3	Department of the Army Reuse of Army Computer Hard Disk Drives, 8 April 2019.
PAM 25-2-8	Department of the Army Cybersecurity: Sanitization of Media, 10 April 2019.
PAM 25-2-16	Department of the Army Communications Security (COMSEC), 8 April 2019.

DRAFT